

DISCRETE FUNCTIONS

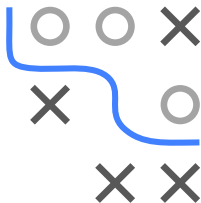
For simplicity, let us consider functions with finite domains first.

Let $\mathcal{X} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$ be a finite set of elements and $\mathcal{H}$ the set of all functions from $\mathcal{X} \rightarrow \mathbb{R}$.

Remark: $\mathcal{X}$ does not mean the training data here but means the “real” domain of the functions.

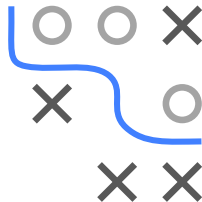
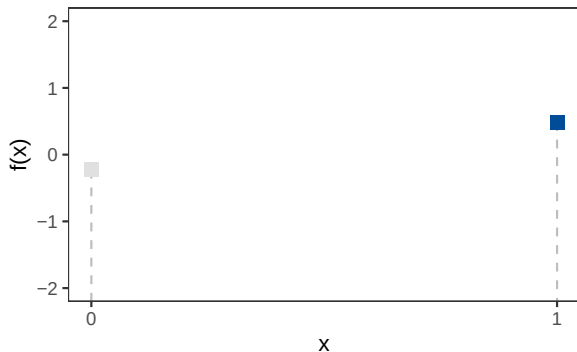
Since the domain of any $f(\cdot) \in \mathcal{H}$ has only n elements, we can represent the function $f(\cdot)$ compactly as a n -dimensional vector

$$\mathbf{f} = \left[f\left(\mathbf{x}^{(1)}\right), \dots, f\left(\mathbf{x}^{(n)}\right) \right].$$



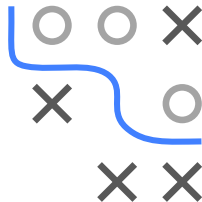
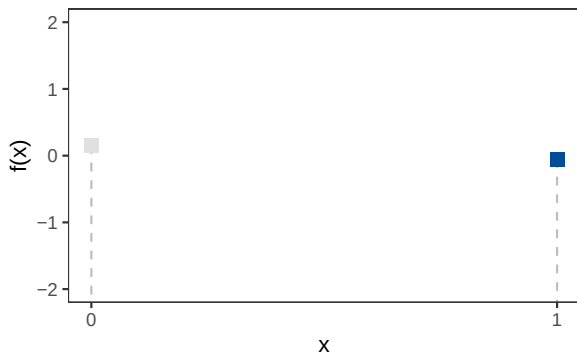
DISCRETE FUNCTIONS

Some examples $f : \mathcal{X} \rightarrow \mathbb{R}$ where $\mathcal{X}$ is univariate and finite:



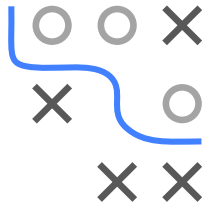
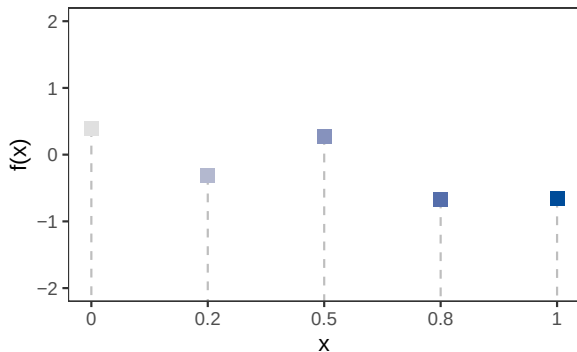
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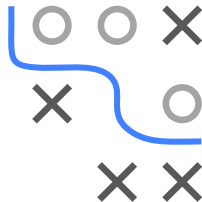
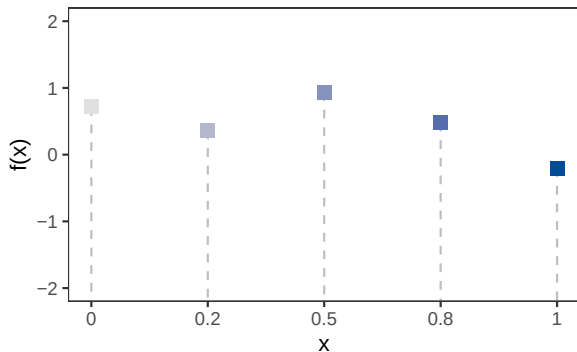
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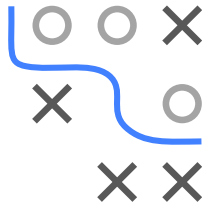
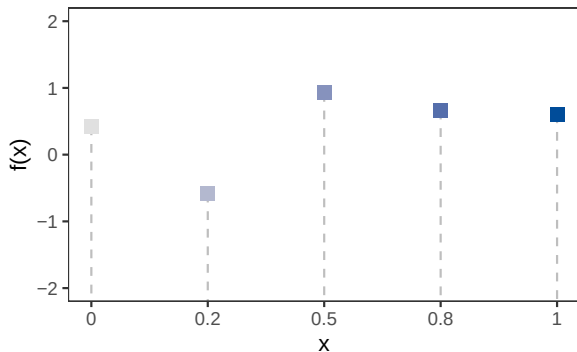
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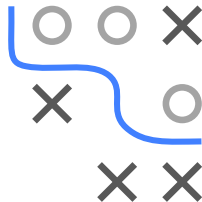
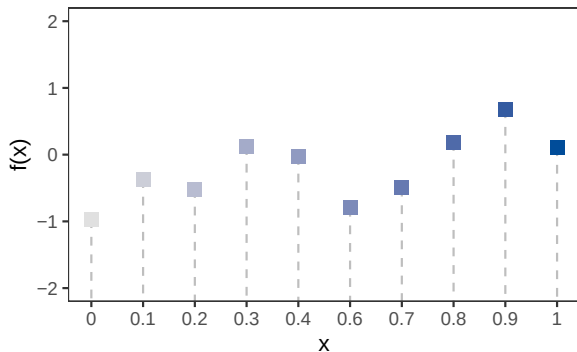
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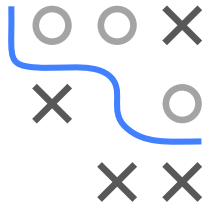
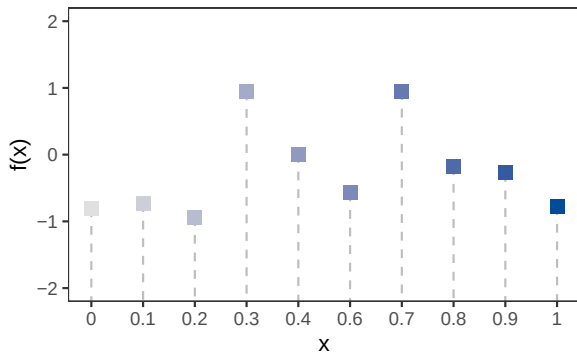
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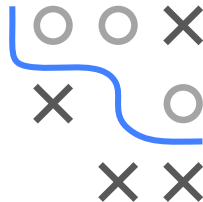
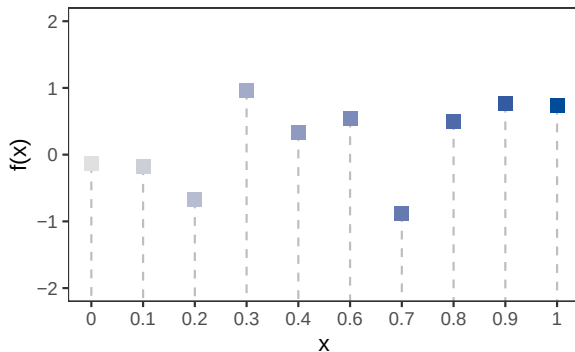
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DISTRIBUTIONS ON DISCRETE FUNCTIONS

One natural way to specify a probability function on a discrete function $f \in \mathcal{H}$ is to use the vector representation

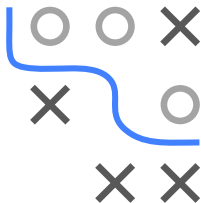
$$\mathbf{f} = \left[f(\mathbf{x}^{(1)}), f(\mathbf{x}^{(2)}), \dots, f(\mathbf{x}^{(n)}) \right]$$

of the function.

Let us see $\mathbf{f}$ as a n -dimensional random variable. We will further assume the following normal distribution:

$$\mathbf{f} \sim \mathcal{N}(\mathbf{m}, \mathbf{K}).$$

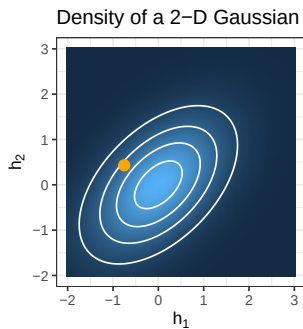
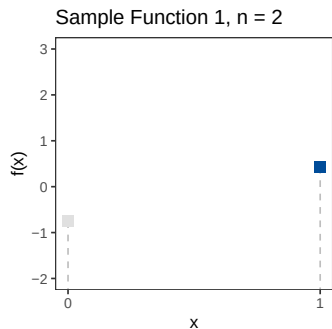
Note: For now, we set $\mathbf{m} = \mathbf{0}$ and take the covariance matrix $\mathbf{K}$ as given. We will see later how they are chosen / estimated.



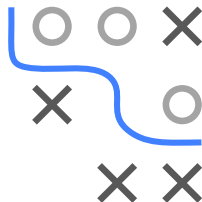
DISCRETE FUNCTIONS

Let $f : \mathcal{X} \rightarrow \mathbb{R}$. Sample functions by sampling from a two-dimensional normal variable.

$$\mathbf{f} = [f(1), f(2)] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$



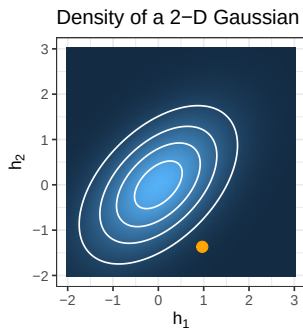
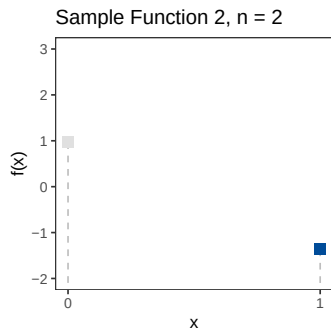
In this example, $\mathbf{m} = (0, 0)$ and $\mathbf{K} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$.



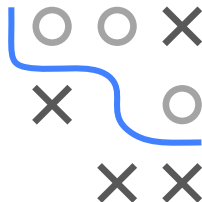
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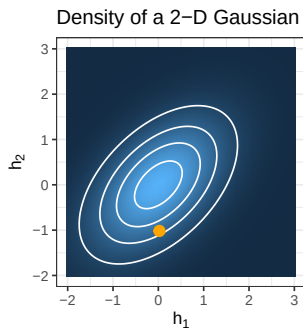
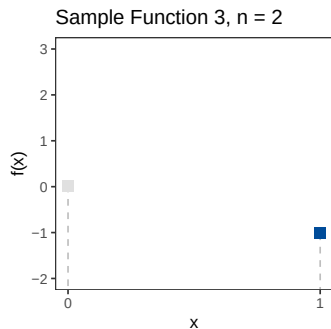
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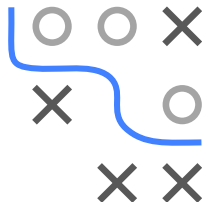
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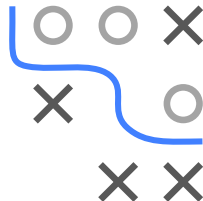
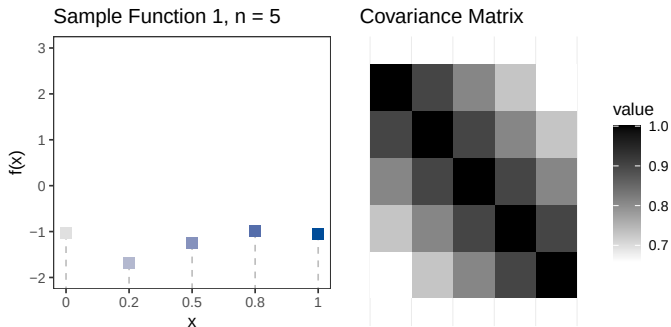
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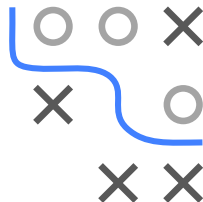
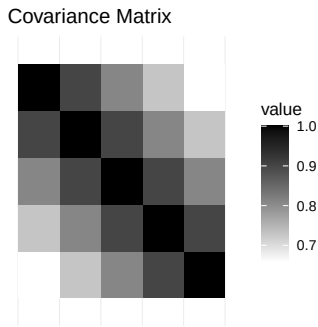
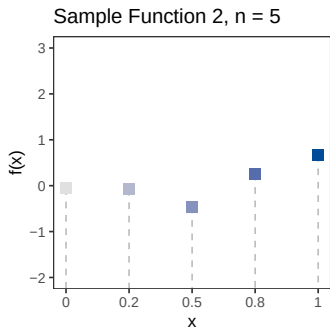
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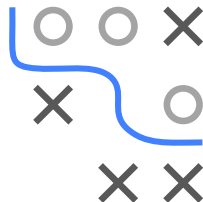
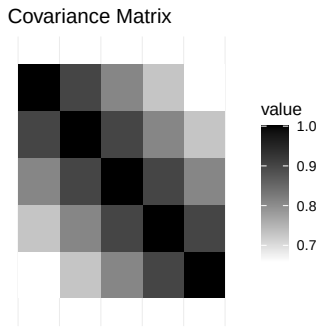
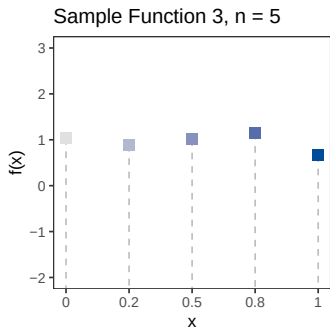
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ROLE OF THE COVARIANCE FUNCTION

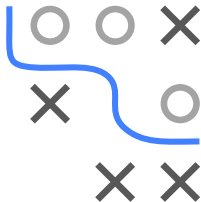
- “Meaningful” functions (on a numeric space $\mathcal{X}$) may be characterized by a spatial property:

If two points $\mathbf{x}^{(i)}, \mathbf{x}^{(j)}$ are close in $\mathcal{X}$ -space, their function values $f(\mathbf{x}^{(i)}), f(\mathbf{x}^{(j)})$ should be close in $\mathcal{Y}$ -space.

In other words: If they are close in $\mathcal{X}$ -space, their functions values should be **correlated**!

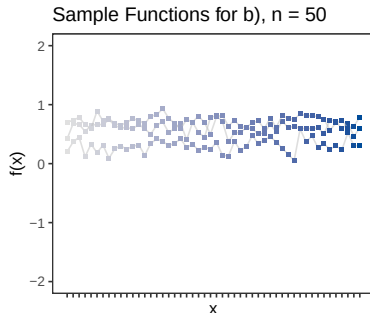
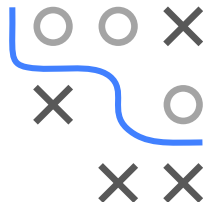
- We can enforce that by choosing a covariance function with

K_{ij} high, if $\mathbf{x}^{(i)}, \mathbf{x}^{(j)}$ close.



ROLE OF THE COVARIANCE FUNCTION

b) Correlation almost 1: $\mathbf{K} = \begin{pmatrix} 1 & 0.99 & \dots & 0.99 \\ 0.99 & 1 & \dots & 0.99 \\ 0.99 & 0.99 & \ddots & 0.99 \\ 0.99 & \dots & 0.99 & 1 \end{pmatrix}.$



Points are highly correlated. Functions become very smooth and flat.